

so you only have to write blocks of code that perform certain actions once, while using that code as many times as you need. You'll then extend this concept to more complicated behavior with classes, making fairly simple programs respond to a variety of situations. You'll learn to write programs that handle common errors gracefully. After working through each of these basic concepts, you'll write a number of increasingly complex programs using what you've learned. Finally, you'll take your first step toward intermediate programming by learning how to write tests for your code, so you can develop your programs further without worrying about introducing bugs. All the information in Part I will prepare you for taking on larger, more complex projects.

In Part II, you'll apply what you learned in Part I to three projects. You can do any or all of these projects, in whichever order works best for you. In the first project, in Chapters 12–14, you'll create a *Space Invaders*-style shooting game called *Alien Invasion*, which includes several increasingly difficult levels of game play. After you've completed this project, you should be well on your way to being able to develop your own 2D games. Even if you don't aspire to become a game programmer, working through this project is an enjoyable way to tie together much of what you'll learn in Part I.

The second project, in Chapters 15–17, introduces you to data visualization. Data scientists use a variety of visualization techniques to help make sense of the vast amount of information available to them. You'll work with datasets that you generate through code, datasets that you download from online sources, and datasets your programs download automatically. After you've completed this project, you'll be able to write programs that sift through large datasets and make visual representations of many different kinds of information.

In the third project, in Chapters 18–20, you'll build a small web application called Learning Log. This project allows you to keep an organized journal of information you've learned about a specific topic. You'll be able to keep separate logs for different topics and allow others to create an account and start their own journals. You'll also learn how to deploy your project so anyone can access it online, from anywhere in the world.

Online Resources

No Starch Press has more information about this book available online at <https://nostarch.com/python-crash-course-3rd-edition>.

I also maintain an extensive set of supplementary resources at https://ehmatthes.github.io/pcc_3e. These resources include the following:

Setup instructions The setup instructions online are identical to what's in the book, but they include active links you can click for all the different steps. If you're having any setup issues, refer to this resource.

Updates Python, like all languages, is constantly evolving. I maintain a thorough set of updates, so if anything isn't working, check here to see whether instructions have changed.